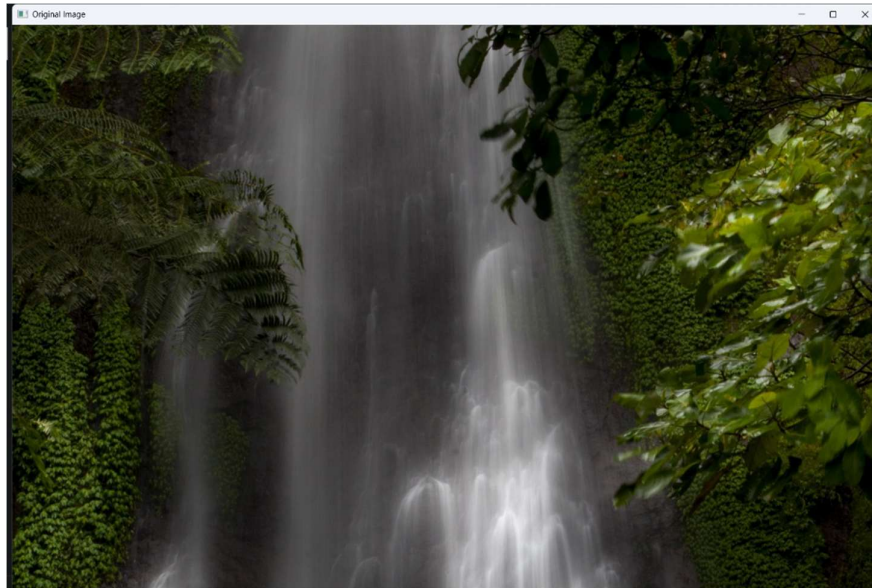


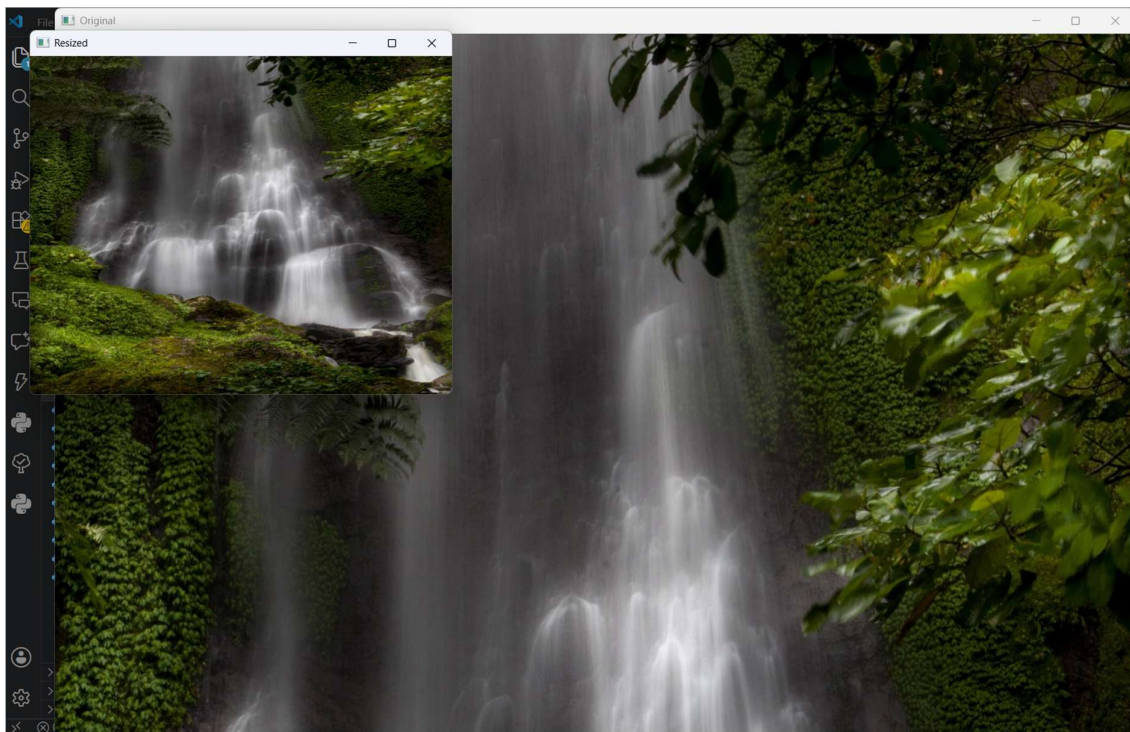
01_read_display_save.py

```
01_read_display_save.py > ...  
1  import cv2  
2  
3  img = cv2.imread("images/sample.jpg")  
4  
5  cv2.imshow("Original Image", img)  
6  
7  cv2.imwrite("output/saved_image.jpg", img)  
8  
9  cv2.waitKey(0)  
10  
11 cv2.destroyAllWindows()
```



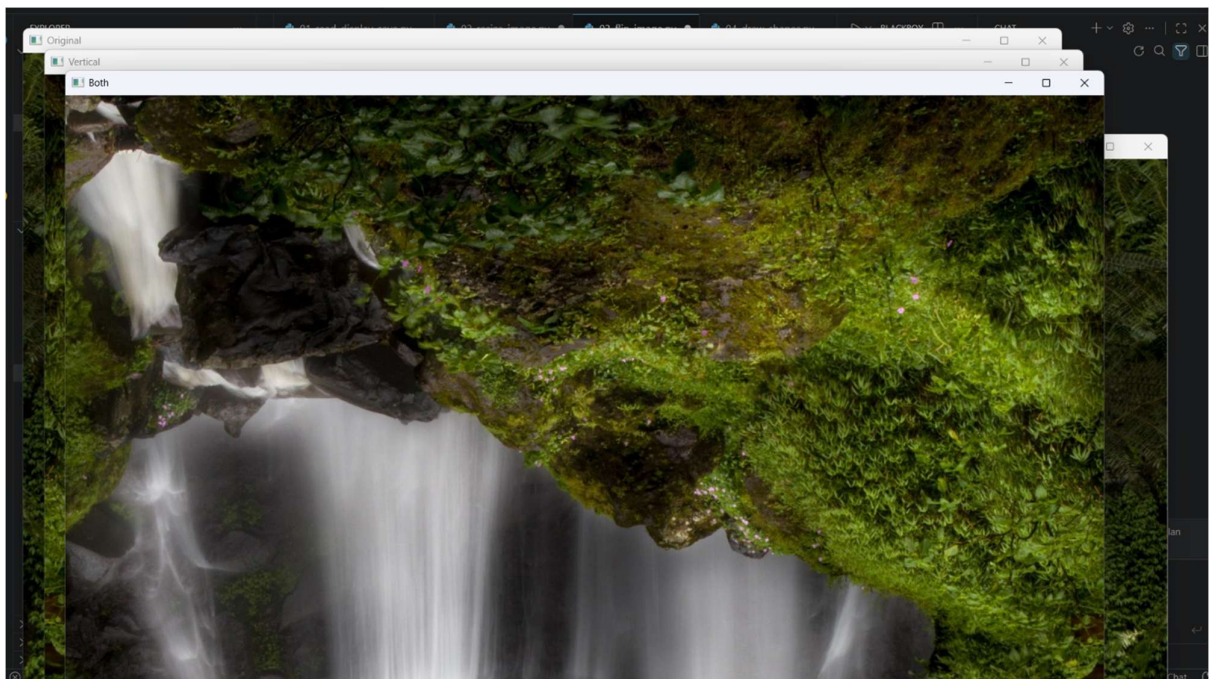
python 02_resize_image.py

```
02_resize_image.py > ...
1  import cv2
2
3  img = cv2.imread("images/sample.jpg")
4
5  resized = cv2.resize(img, (500,400))
6
7  cv2.imshow("Original", img)
8  cv2.imshow("Resized", resized)
9
10 cv2.imwrite("output/resized.jpg", resized)
11
12 cv2.waitKey(0)
13 cv2.destroyAllWindows()
```



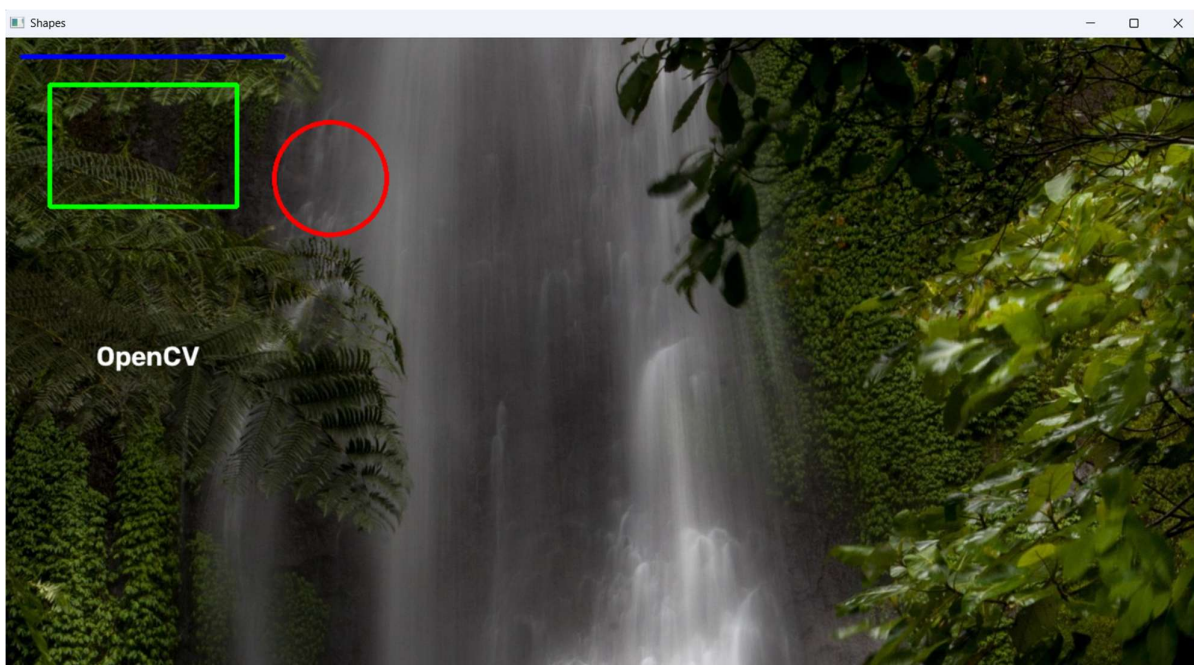
python 03_flip_image.py

```
03_flip_image.py > ...
1  import cv2
2
3  img = cv2.imread("images/sample.jpg")
4
5  horizontal = cv2.flip(img,1)
6  vertical = cv2.flip(img,0)
7  both = cv2.flip(img,-1)
8
9  cv2.imshow("Original",img)
10 cv2.imshow("Horizontal",horizontal)
11 cv2.imshow("Vertical",vertical)
12 cv2.imshow("Both",both)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
```



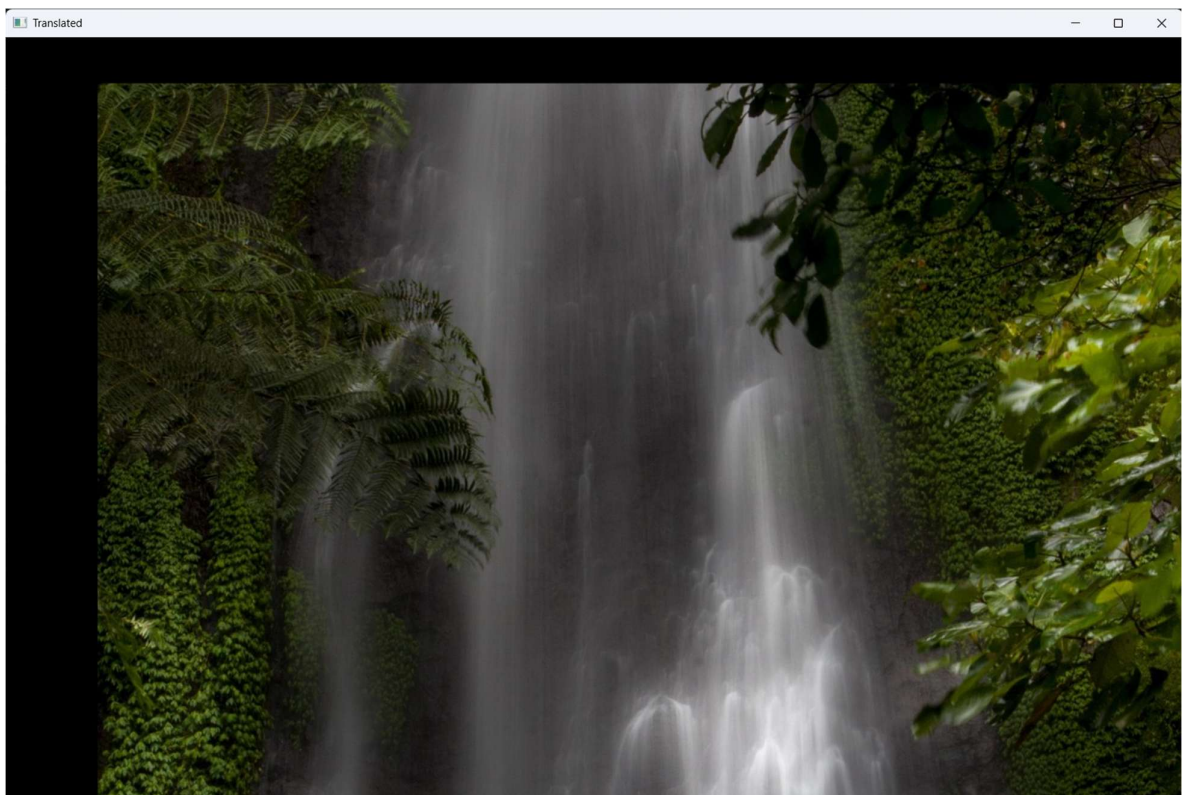
python 04_draw_shapes.py

```
04_draw_shapes.py > ...
1  import cv2
2
3  img = cv2.imread("images/sample.jpg")
4
5  cv2.line(img, (20,20), (300,20), (255,0,0),3)
6
7  cv2.rectangle(img, (50,50), (250,180), (0,255,0),3)
8
9  cv2.circle(img, (350,150), 60, (0,0,255),3)
10
11 cv2.putText(img, "OpenCV", (100,350),
12 cv2.FONT_HERSHEY_SIMPLEX,
13 1, (255,255,255), 2)
14
15 cv2.imshow("Shapes", img)
16
17 cv2.waitKey(0)
18 cv2.destroyAllWindows()
```



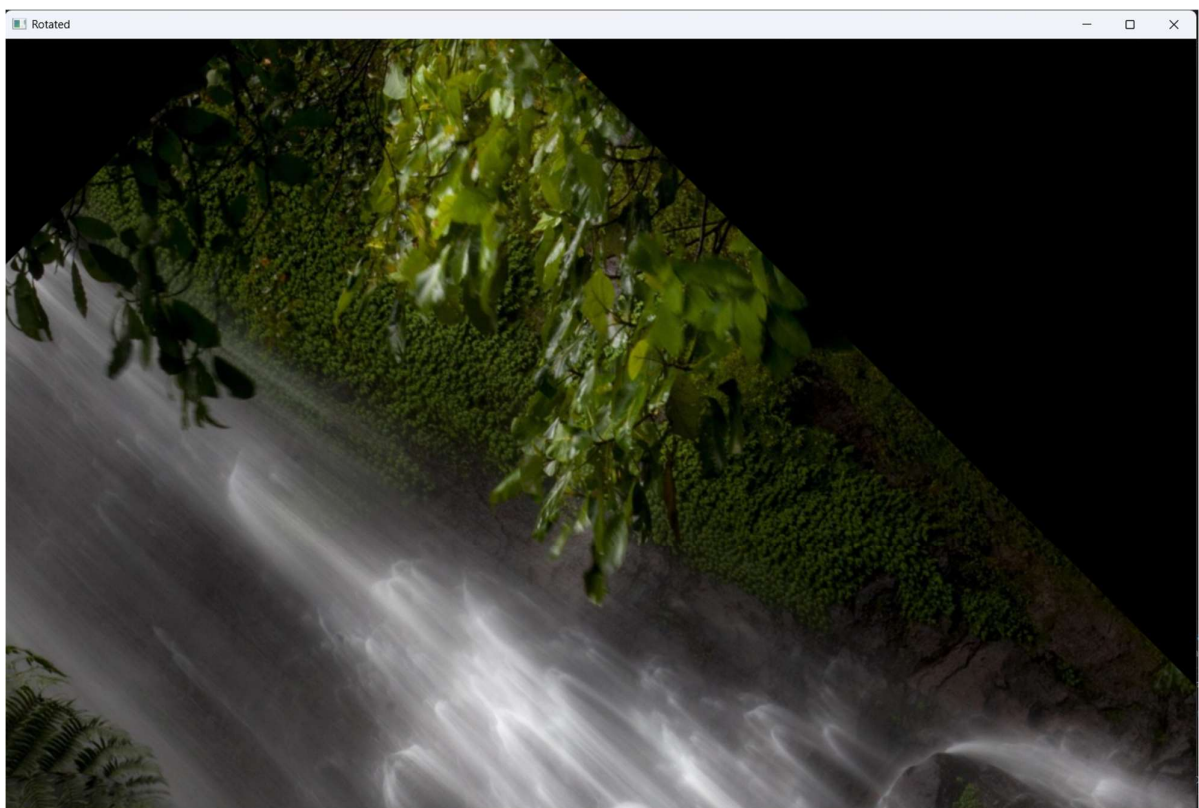
05_translation.py

```
05_translation.py > ...
1  import cv2
2  import numpy as np
3
4  img=cv2.imread("images/sample.jpg")
5
6  rows,cols=img.shape[:2]
7
8  M=np.float32([[1,0,100],[0,1,50]])
9
10 shift=cv2.warpAffine(img,M,(cols,rows))
11
12 cv2.imshow("Translated",shift)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
```



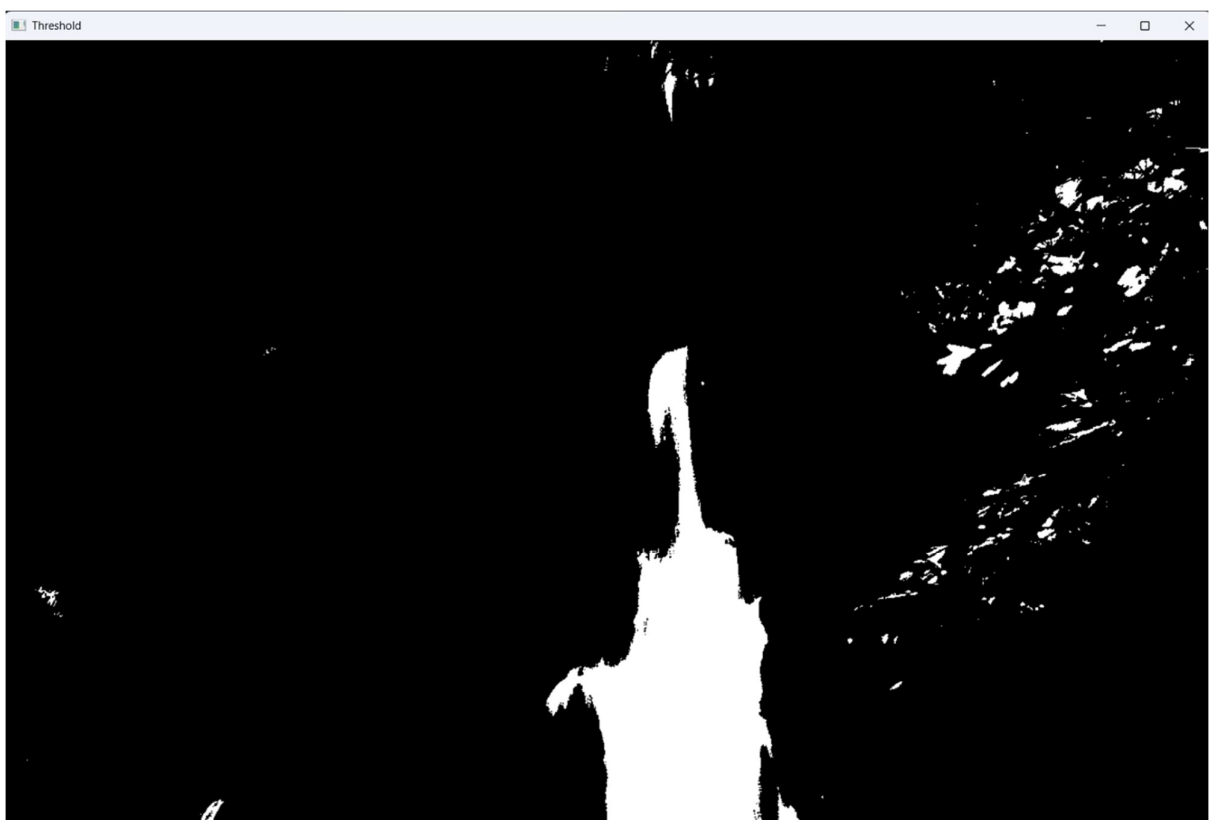
06_rotation.py

```
06_rotation.py > ...
1  import cv2
2
3  img=cv2.imread("images/sample.jpg")
4
5  rows,cols=img.shape[:2]
6
7  M=cv2.getRotationMatrix2D((cols/2,rows/2),45,1)
8
9  rotate=cv2.warpAffine(img,M,(cols,rows))
10
11  cv2.imshow("Rotated",rotate)
12
13  cv2.waitKey(0)
14  cv2.destroyAllWindows()
```



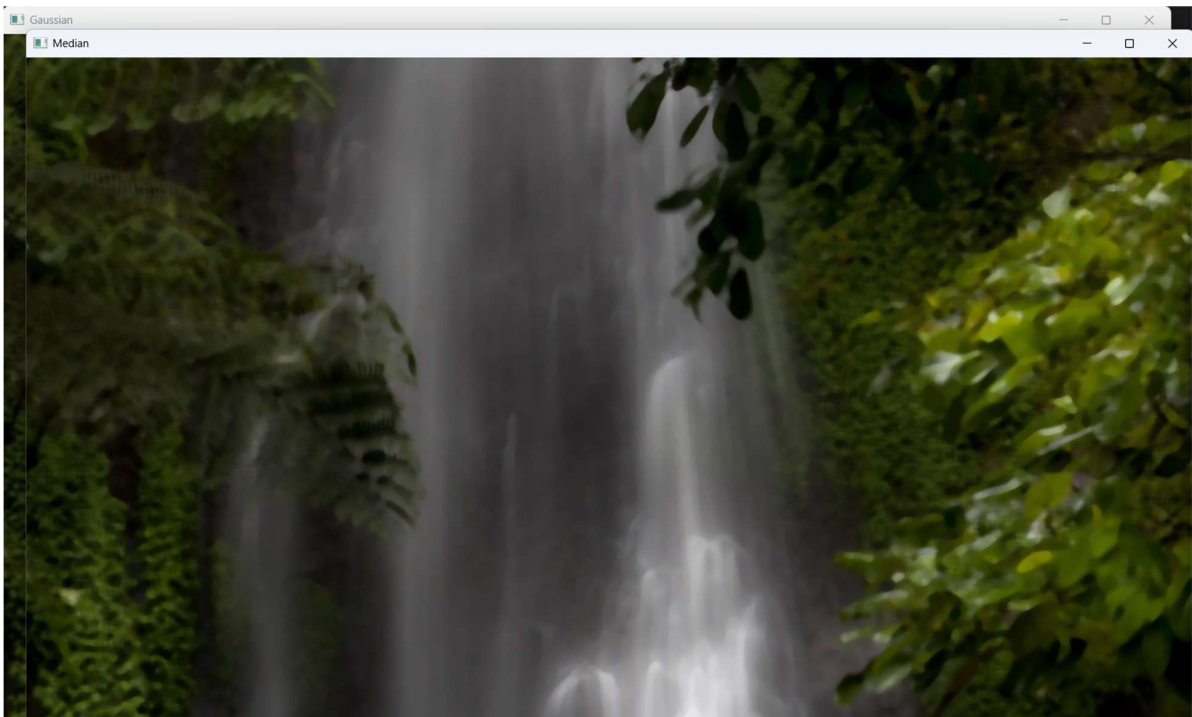
python 07_threshold.py

```
07_threshold.py > ...
1  import cv2
2
3  img=cv2.imread("images/sample.jpg",0)
4
5  _,th=cv2.threshold(img,127,255,cv2.THRESH_BINARY)
6
7  cv2.imshow("Threshold",th)
8
9  cv2.waitKey(0)
10 cv2.destroyAllWindows()
```



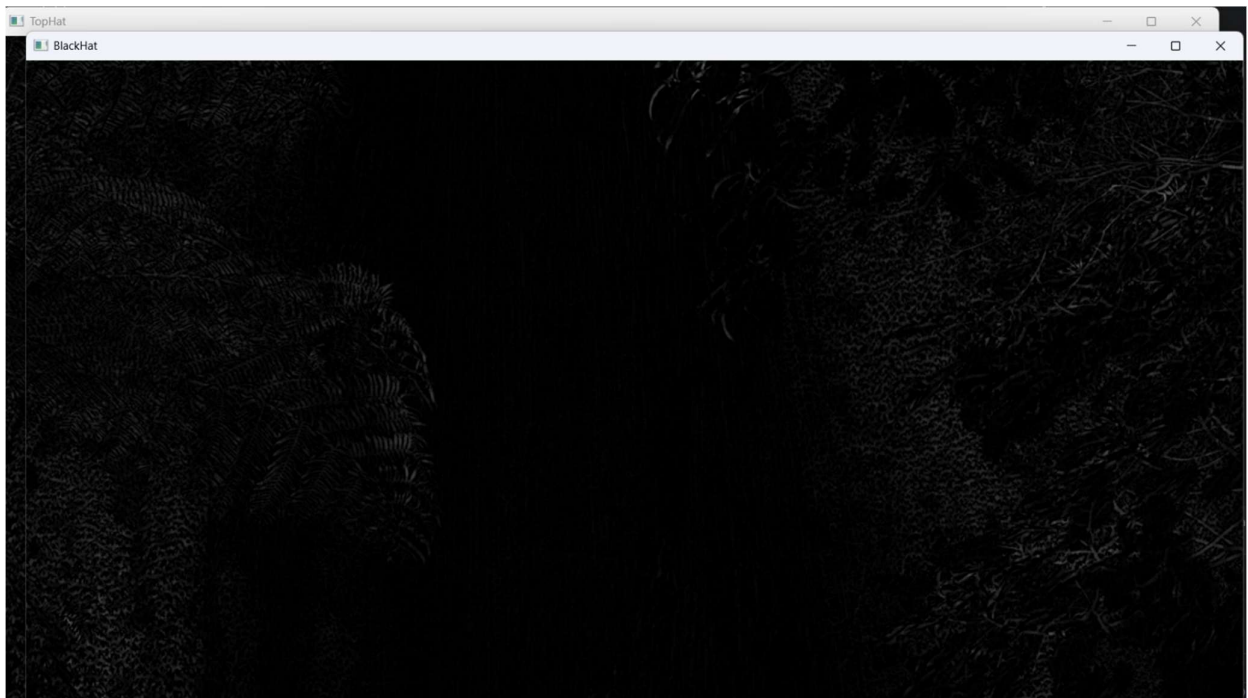
08_blur.py

```
08_blur.py > ...  
1  import cv2  
2  
3  img=cv2.imread("images/sample.jpg")  
4  
5  gaussian=cv2.GaussianBlur(img,(9,9),0)  
6  
7  median=cv2.medianBlur(img,9)  
8  
9  cv2.imshow("Gaussian",gaussian)  
10 cv2.imshow("Median",median)  
11  
12 cv2.waitKey(0)  
13 cv2.destroyAllWindows()
```



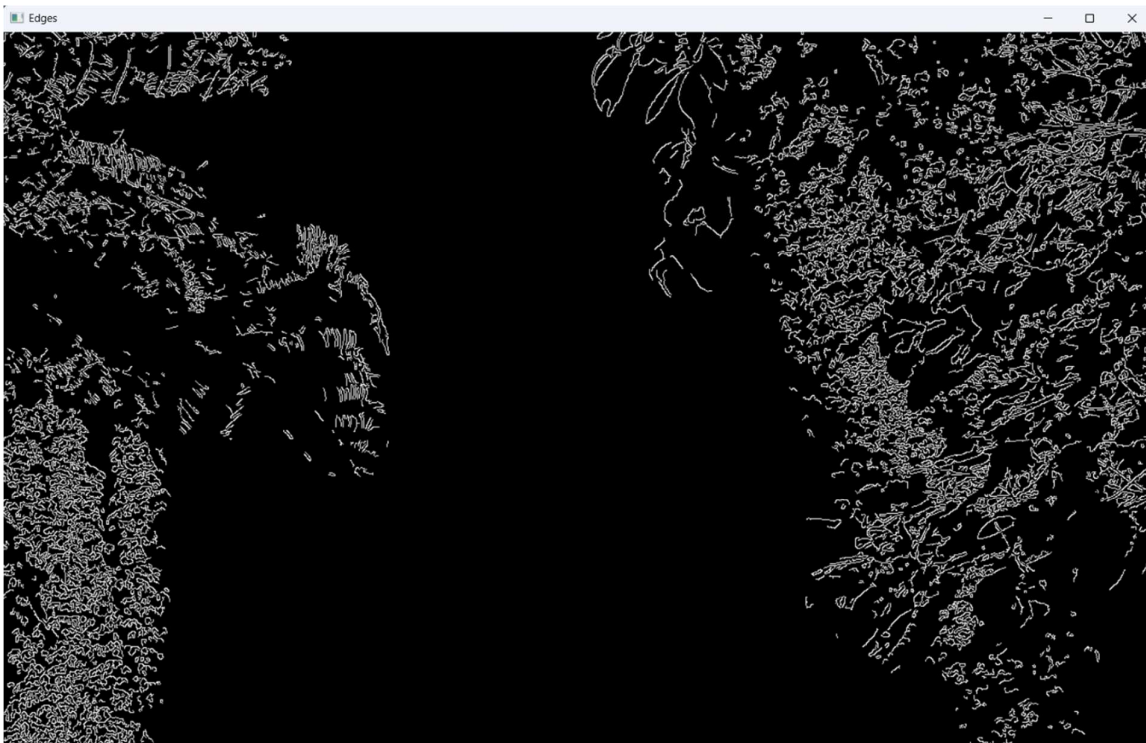
09_morphology.py

```
09_morphology.py > ...
1  import cv2
2  import numpy as np
3
4  img=cv2.imread("images/sample.jpg",0)
5
6  kernel=np.ones((5,5),np.uint8)
7
8  tophat=cv2.morphologyEx(img,cv2.MORPH_TOPHAT,kernel)
9
10 blackhat=cv2.morphologyEx(img,cv2.MORPH_BLACKHAT,kernel)
11
12 cv2.imshow("TopHat",tophat)
13 cv2.imshow("BlackHat",blackhat)
14
15 cv2.waitKey(0)
16 cv2.destroyAllWindows()
```



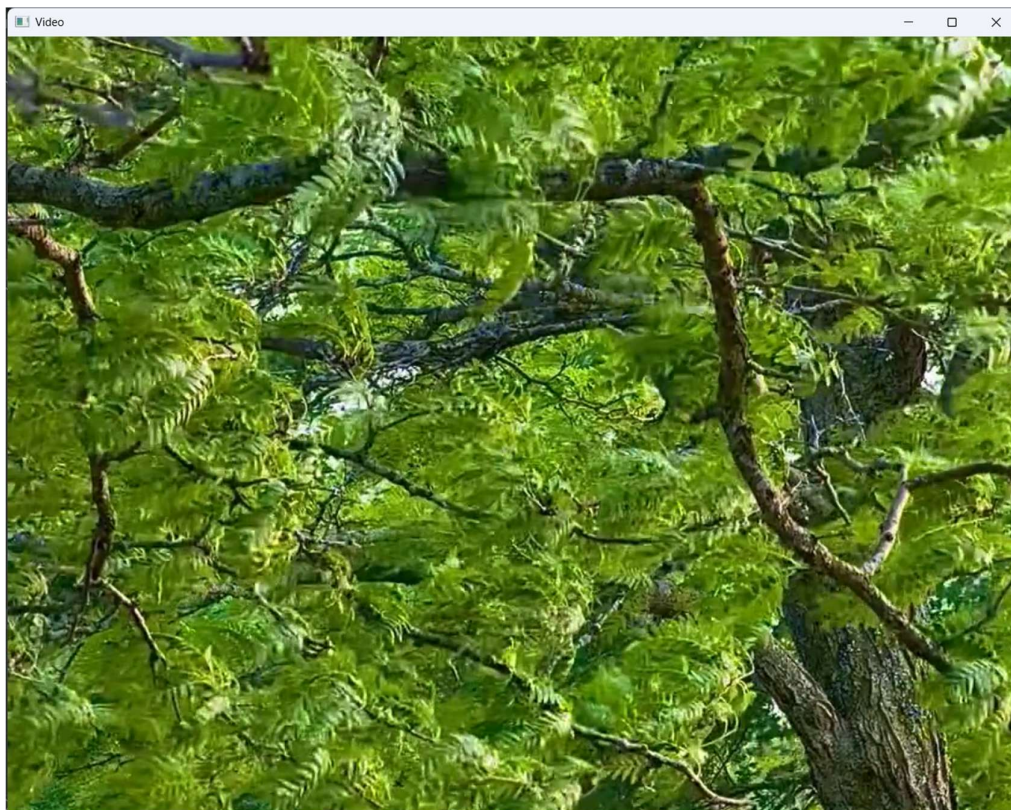
10_edge_detection.py

```
10_edge_detection.py > ...  
1  import cv2  
2  
3  img=cv2.imread("images/sample.jpg",0)  
4  
5  edge=cv2.Canny(img,100,200)  
6  
7  cv2.imshow("Edges",edge)  
8  
9  cv2.waitKey(0)  
10 cv2.destroyAllWindows()
```



11_video.py

```
11_video.py > ...
1  import cv2
2
3  cap=cv2.VideoCapture("videos/sample.mp4")
4
5  while True:
6
7      ret,frame=cap.read()
8
9      if not ret:
10         break
11
12         cv2.imshow("Video",frame)
13
14         if cv2.waitKey(25)&0xFF==ord('q'):
15             break
16
17 cap.release()
18 cv2.destroyAllWindows()
```



12_webcam.py

```
12_webcam.py > ...
1  import cv2
2
3  cap=cv2.VideoCapture(0)
4
5  while True:
6
7      ret,frame=cap.read()
8
9      if not ret:
10         break
11
12         cv2.imshow("Webcam",frame)
13
14         if cv2.waitKey(1)&0xFF==ord('q'):
15             break
16
17 cap.release()
18 cv2.destroyAllWindows()
```

